Properties and Applications of Cast Iron

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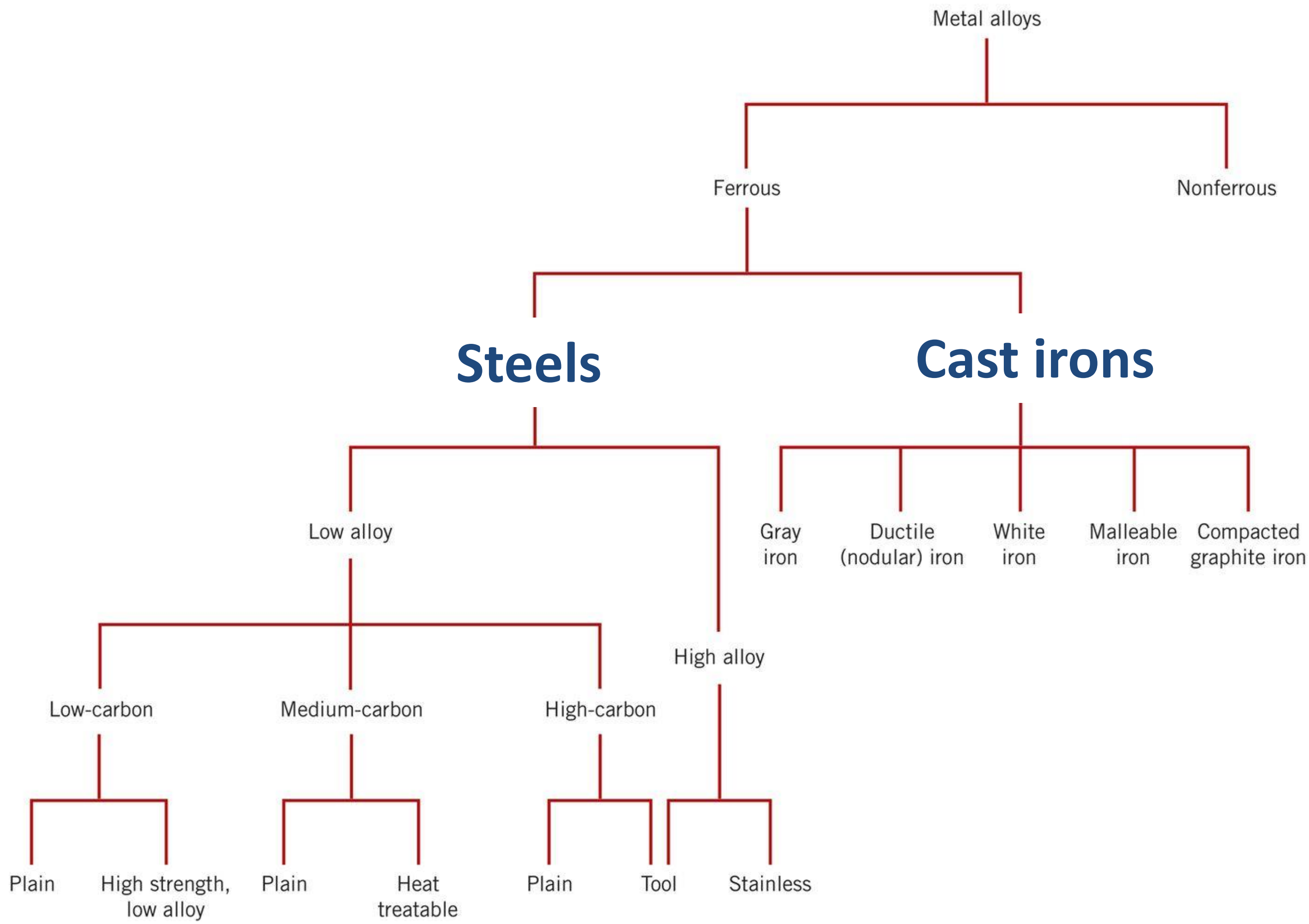
Ferrous Metals

Ferrous Metals are sub-divided into categories:



Fe – C phase diagram

True equilibrium Fe – C phase diagram → 100% C = graphite

Graphite formation promoted by

- Si > 1 wt%
- slow cooling

Instead of Fe_3C

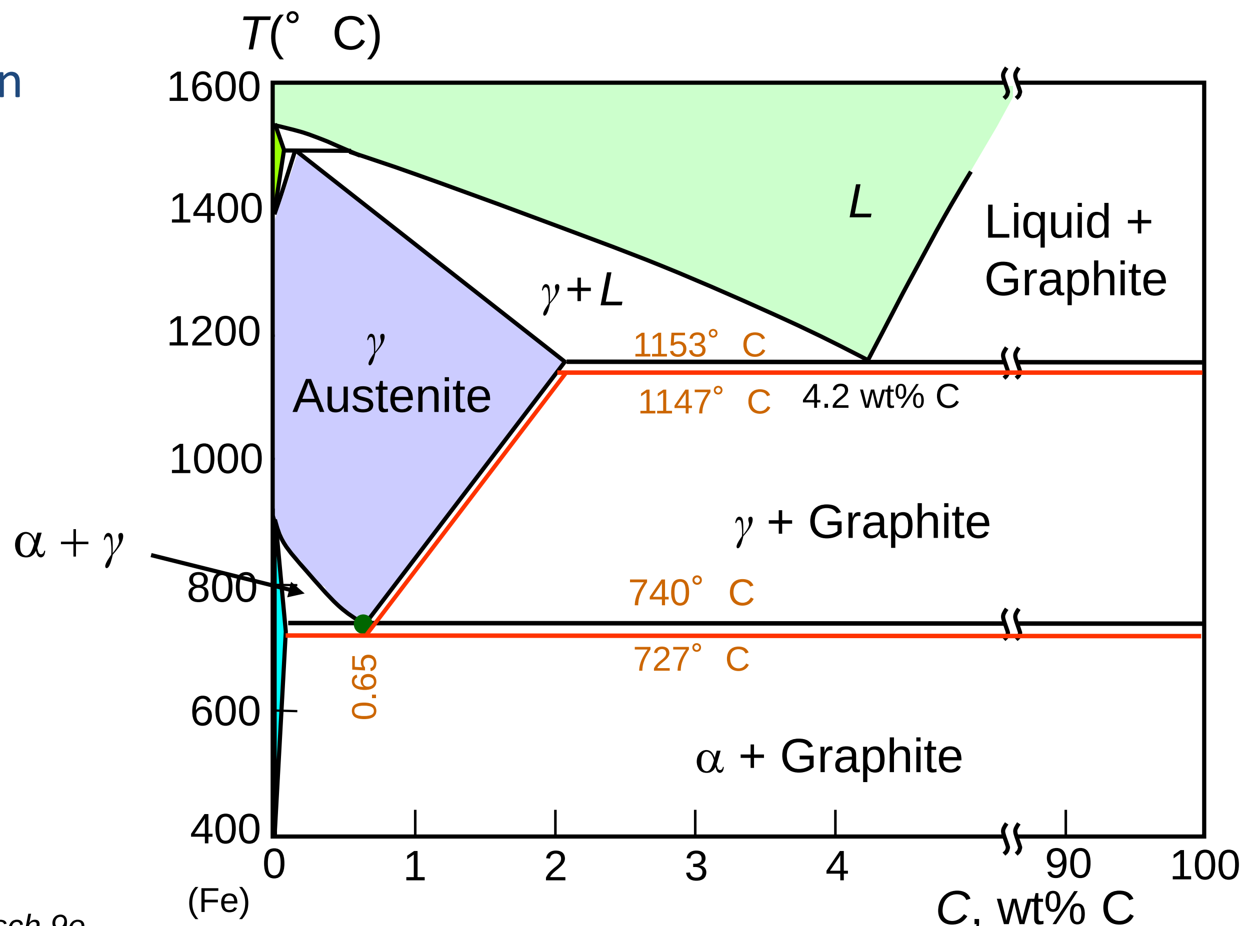
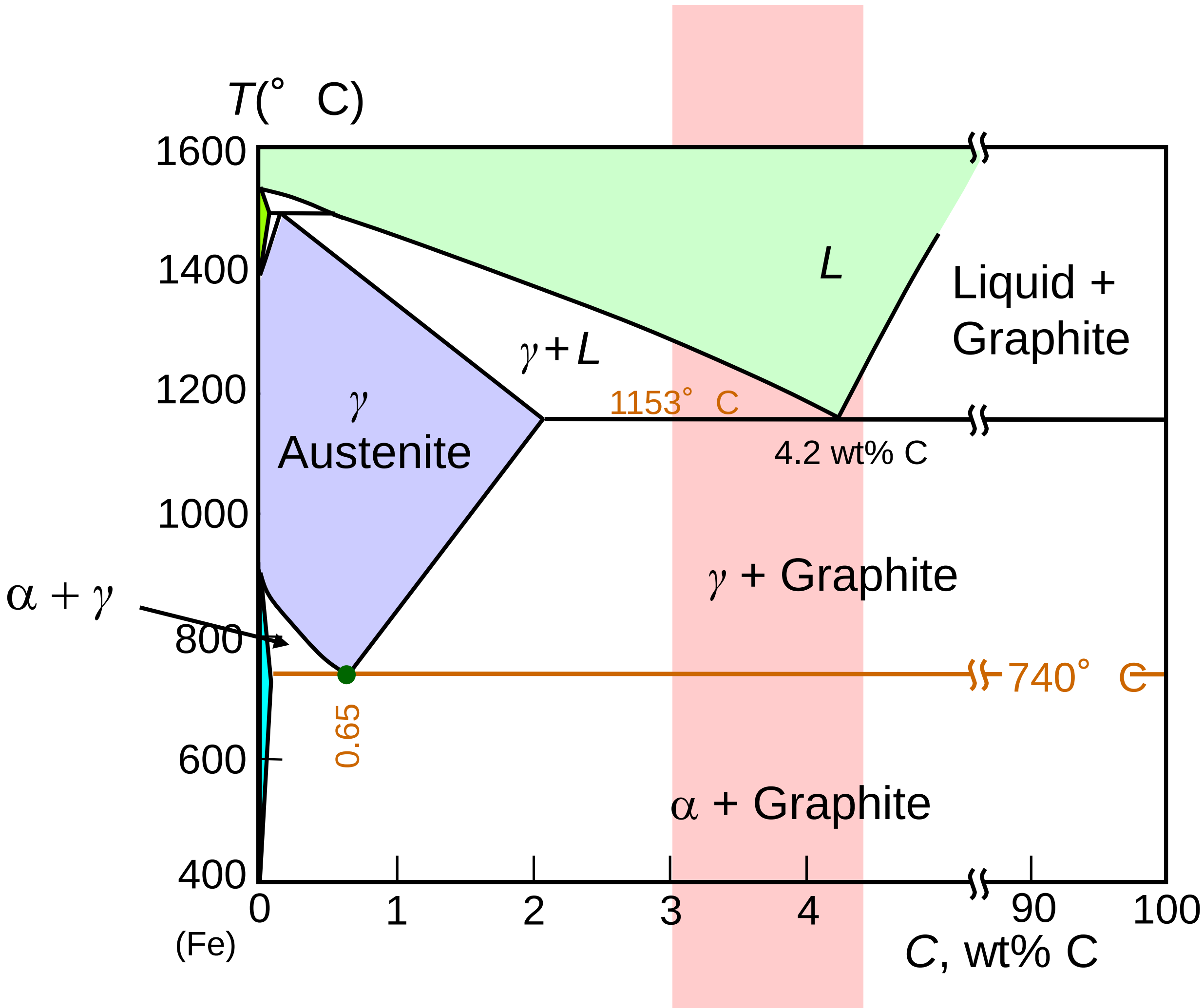


Fig. 13.2, *Callister & Rethwisch 9e.*
[Adapted from *Binary Alloy Phase Diagrams*, T. B. Massalski (Editor-in-Chief), 1990. Reprinted by permission of ASM International, Materials Park, OH.]

Fe – C phase diagram

Cast irons = Carbon > 2.1wt% (typically 3-4.5wt%)



Fe – C phase diagram

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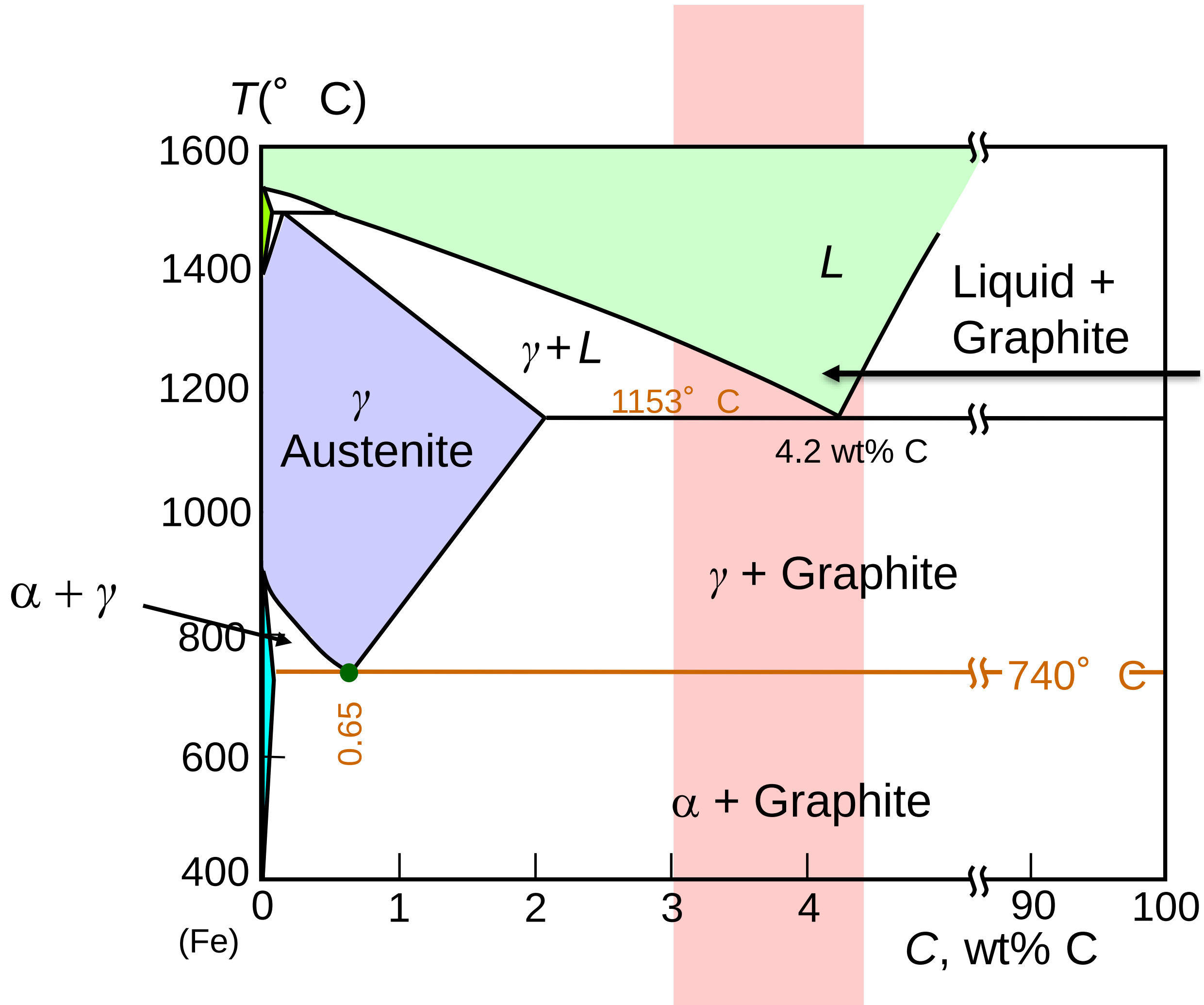


Low melting point 1150-1300°C

Cast Iron

- Ferrous alloys with > 2.1 wt% C
  - more commonly 3 - 4.5 wt% C

- Low melting ~1300°C – solidifies quickly on cooling

  = relatively easy to **cast** (earliest iron materials)

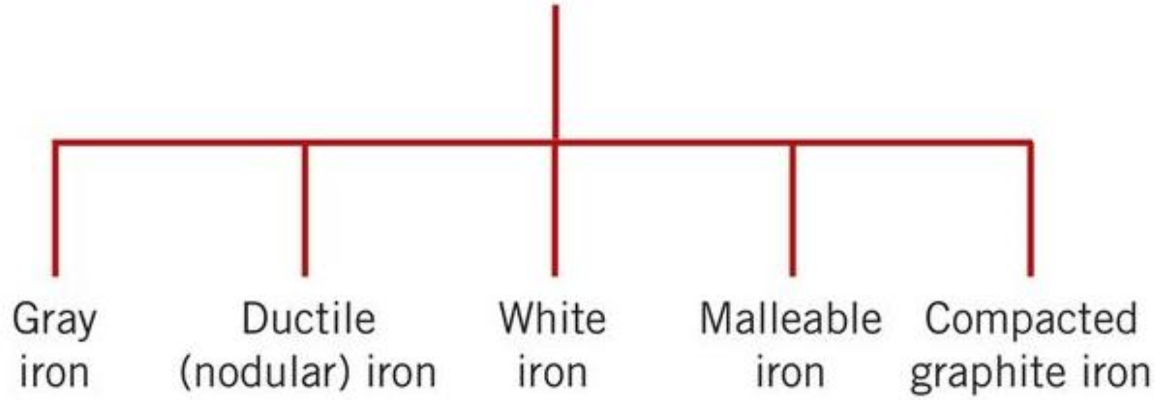
- Cementite decomposes to ferrite + graphite

  $Fe_3C \rightarrow 3\ Fe\ (\alpha) + C\ (\text{graphite})$

  - generally a slow process, promoted by > 1wt% Si

Types of Cast Iron

There are 5 types of **cast iron**



- The type depends on the appearance of the microstructure
- The form that the graphite takes changes the properties
- In general, cast irons are often **brittle**

Gray Cast Iron

Microstructure:

- carbon = 2.5 – 4.0 wt%; silicon = 1.0 – 3.0 wt%
- graphite in the form of **flakes** in a pearlite or α-ferrite matrix
- graphite weak and brittle in tension
- flakes act as stress concentration for brittle fracture
- Fracture surface reveals graphite, giving the gray appearance (so the name)

Applications:

- stronger in compression
- excellent vibrational dampening
- wear resistant
- used in heavy machinery

Figs. 13.3(a), *Callister & Rethwisch 9e.*
[Courtesy of C. H. Brady and L. C. Smith, National Bureau of Standards, Washington, DC (now the National Institute of Standards and Technology, Gaithersburg, MD]

Ductile Iron

Microstructure:

- add Mg and/or Ce
- graphite as nodules/spherelike not flakes
- matrix often α + Fe_3C (pearlite)

  – instead of ferrite α (ferrite)

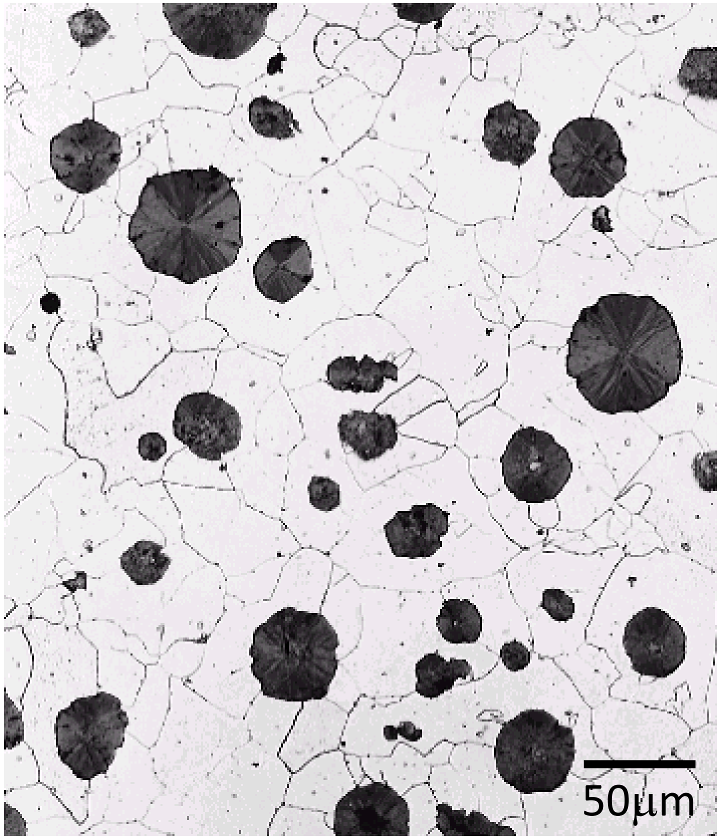
  depend on the heat treatment/cooling rate

  pearlite = stronger but less ductile

- ferrite matrix = ductile material
- pearlite matrix = higher strength

Applications:

- castings of valves and pumps
- crankshafts and gears
- machinery



Fully annealed at 700°C = ferrite matrix

Figs. 13.3 (b), *Callister & Rethwisch 9e.* [Courtesy of C. H. Brady and L. C. Smith, National Bureau of Standards, Washington, DC (now the National Institute of Standards and Technology, Gaithersburg, MD]

White Iron

Microstructure:

- low Si content < 1 wt% + fast cooling
- cementite microstructure (not graphite)

  = ferrite + pearlite
- very hard and brittle
- fracture surface appears white (so the name)
- slower cooling forms gray iron

Applications:

- rollers for materials processing

  hard surface (fast cooling)

  more ductile inside (slower cooling)

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Figs. 13.3(c), *Callister & Rethwisch 9e.*

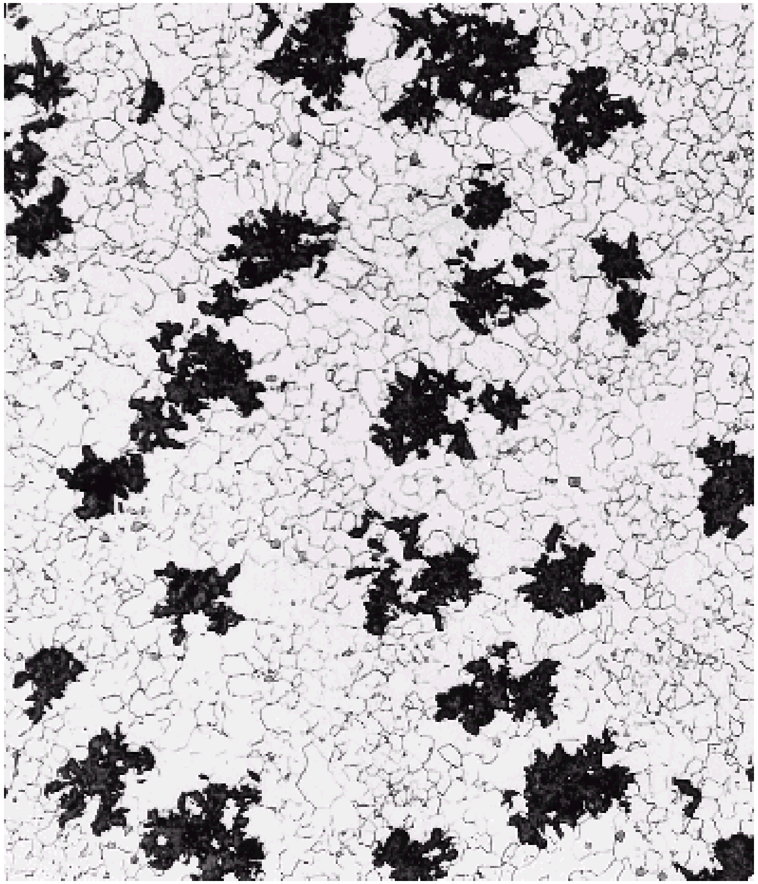
Malleable Iron

Microstructure:

- heat treat white iron at 800-900°C
- pearlite decomposes to graphite
- microstructure is then graphite in form of rosettes in ferrite matrix
- similar to nodular iron
- reasonably strong and ductile

Applications:

- casings for automotive
- flanges and pipe fittings
- rail and marine engineering



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100μm

Figs. 13.3 (d), *Callister & Rethwisch 9e.*

Compacted Graphite Iron

Microstructure:

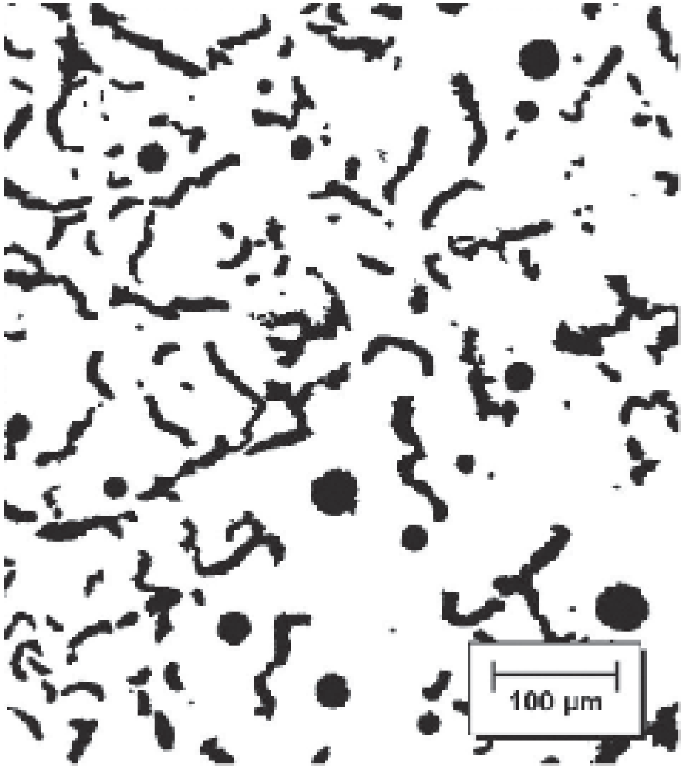
- Si content between 1.7 and 3.0 1 wt%

carbon between 3.1 and 4 wt%

- graphite in 'worm-like' (or vermicular) shape
- less brittle than gray iron
- improved thermal conductivity
- good resistance to thermal shock
- lower oxidation at elevated temperatures

Applications:

- diesel engine castings
- brake disc for high speed train
- flywheels



Courtesy of Sinter-Cast, Ltd.

Fig. 13.3(e), *Callister & Rethwisch 9e.*

Properties of Cast Iron

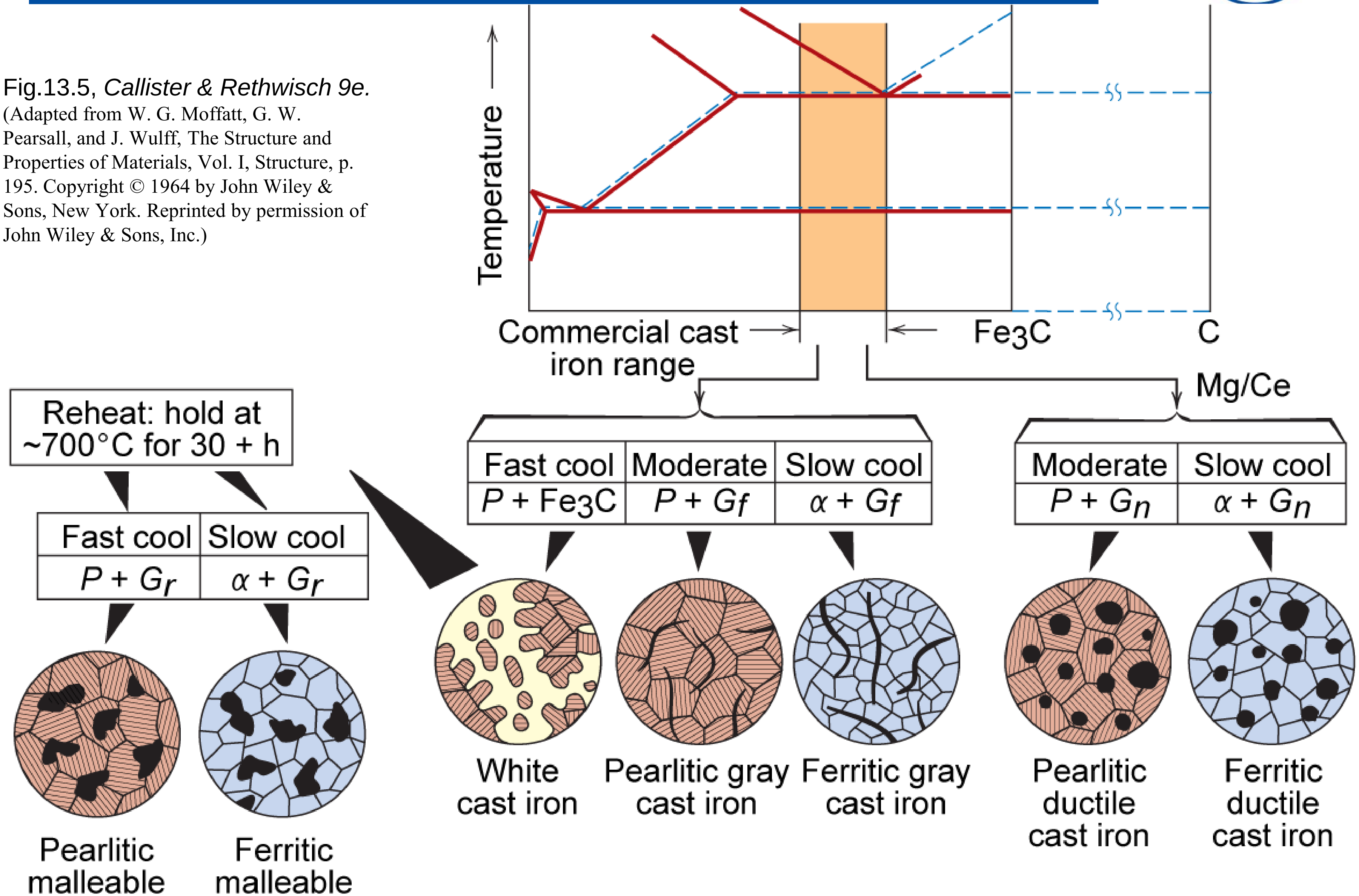
Elastic modulus = 170 GPa (Fe 15% graphite)

Type of cast iron	Microstructure	σ_{UTS} / MPa	σ_y / MPa	EL / %
Gray	ferrite + pearlite ferrite + pearlite pearlite	124 173 276	brittle	
Ductile (nodular)	ferrite pearlite tempered martensite	414 689 827	276 483 621	18 3 2
Malleable iron	ferrite ferrite + pearlite	345 448	224 310	10 6
Compacted graphite	ferrite pearlite	250 450	175 315	3 1

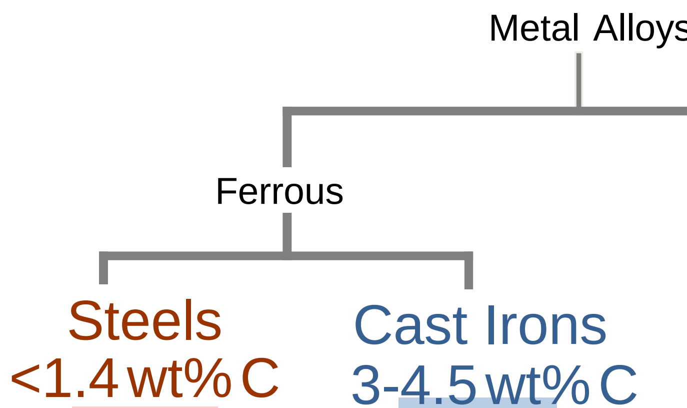
Range of strength possible – mostly low ductility
Material good for casting process

Production of Cast Irons

Fig.13.5, *Callister & Rethwisch 9e.*
(Adapted from W. G. Moffatt, G. W. Pearsall, and J. Wulff, The Structure and Properties of Materials, Vol. I, Structure, p. 195. Copyright © 1964 by John Wiley & Sons, New York. Reprinted by permission of John Wiley & Sons, Inc.)



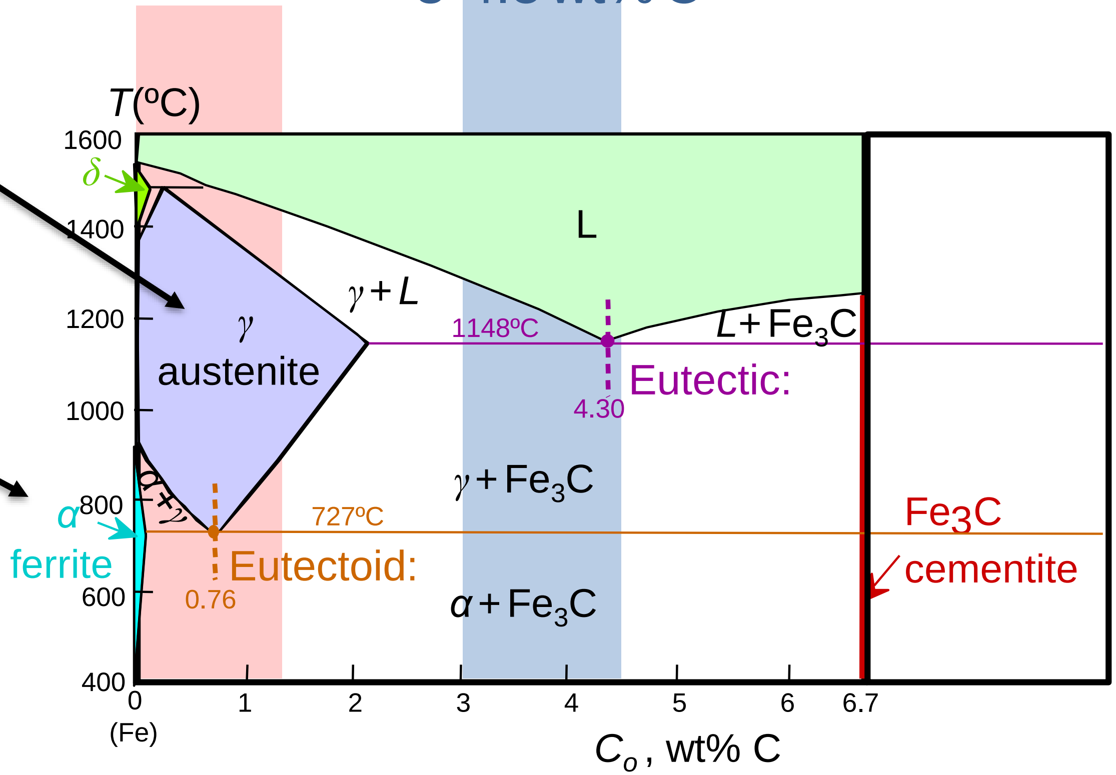
Summary of ferrous alloys



Adapted from Fig. 13.1, *Callister & Rethwisch 9e.*

γ Austenite
= fcc metal
non-magnetic

α Ferrite
= bcc metal
magnetic



100% C
= graphite

Summary of ferrous alloys

Why are they so widely used?

- Good properties
- Many years experience
- Abundant element on earth

Advantages:

- Wide range of properties
- Stiff, strong and ductile
- Cheap to produce on large scale

Disadvantages:

- Corrosion resistance is poor
- Relatively high density
- Poor electrical and thermal conduction